Computer Forensics – Detailed Notes (Session 2)

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1. Computer Forensics Involves

Computer Forensics involves the systematic process of collecting, preserving, analyzing, and presenting digital evidence in a manner acceptable in a court of law.

The main activities are:

1. Preservation
2. Identification
3. Extraction
4. Documentation
5. Interpretation

Real-Time Example

A company's confidential data is leaked.

A forensic investigator:

- Identifies the suspect's laptop.
- Preserves the hard disk.
- Extracts deleted files.
- Documents every action.
- Interprets evidence to determine who leaked the data.

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2. Preservation

Definition

Preservation means protecting digital evidence from modification, deletion, or contamination.

Why Preservation is Important?

Digital evidence can be easily altered.

If evidence changes, it may not be accepted in court.

Activities

- Disconnect affected systems.
- Create forensic image.
- Use write blockers.
- Calculate hash values (MD5/SHA256).

Real-Time Example

Police seize a hacker's laptop.

Instead of analyzing the original disk, they create an exact copy (forensic image).

Original evidence remains untouched.

Example

```
Original Hard Disk

Forensic Image

Analysis Performed on Copy
```

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3. Identification

Definition

Identification is the process of locating potential sources of digital evidence.

Sources of Evidence

- Hard Disk
- SSD
- USB Drive
- Mobile Phone
- Email Account
- Cloud Storage
- Network Logs

Real-Time Example

An employee is suspected of stealing company data.

Investigator identifies:

- Office Laptop
- USB Drive
- Gmail Account
- File Server Logs

as possible evidence sources.

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4. Extraction

Definition

Extraction is the process of retrieving relevant data from digital devices.

Extracted Data

- Deleted Files
- Emails
- Browser History
- System Logs
- Chat Messages
- Images
- Videos

Real-Time Example

A cybercriminal deletes ransomware files after attacking a company.

Using forensic tools, investigators recover deleted malware files from the hard disk.

Tools

Tool	Purpose
FTK Imager	Disk Imaging
Autopsy	File Recovery
EnCase	Evidence Analysis
Volatility	Memory Analysis

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5. Documentation

Definition

Documentation means recording every step performed during the investigation.

Why Important?

- Maintains transparency.
- Ensures legal admissibility.
- Supports court proceedings.

Recorded Information

- Date & Time
- Investigator Name
- Device Details
- Evidence Collected
- Tools Used
- Findings

Real-Time Example

Investigator records:

```
Date: 10-Jan-2026
Time: 10:30 AM
Device: Dell Laptop
Serial No: XYZ123
Action: Disk Image Created
Hash: SHA256 Value
```

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6. Interpretation

Definition

Interpretation means analyzing evidence and determining what happened.

Questions Answered

- Who committed the crime?
- How was the attack performed?
- When did it occur?
- What data was affected?

Real-Time Example

Log analysis reveals:

```
USB Inserted : 2:00 PM
Files Copied : 2:05 PM
User Logout  : 2:10 PM
```

Interpretation:

The employee copied confidential files to a USB drive.

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7. Goals of Forensics Analysis

The primary objectives of forensic analysis are:

1. Recover Deleted Data

Example:

Recover deleted project reports.

2. Identify Attackers

Example:

Trace hacker IP addresses.

3. Determine Attack Method

Example:

Find how malware entered the network.

4. Preserve Evidence

Example:

Create forensic disk images.

5. Support Legal Proceedings

Example:

Submit evidence in court.

6. Prevent Future Incidents

Example:

Identify security weaknesses.

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8. Types of Cyber Forensics Techniques

1. Disk Forensics

Investigates storage devices.

Example

Recover deleted files from HDD.

2. Network Forensics

Analyzes network traffic.

Example

Identify source of DDoS attack.

3. Mobile Forensics

Investigates smartphones.

Example

Recover deleted WhatsApp chats.

4. Memory Forensics

Analyzes RAM contents.

Example

Detect malware running in memory.

5. Email Forensics

Investigates emails.

Example

Analyze phishing emails.

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6. Database Forensics

Investigates database activities.

Example

Track unauthorized changes in banking databases.

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7. Cloud Forensics

Investigates cloud environments.

Example

Analyze stolen data from cloud storage.

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9. Cyber Forensics Procedures

The complete forensic procedure consists of:

Step 1: Preparation

Prepare tools and team.

Step 2: Identification

Locate evidence.

Step 3: Preservation

Protect evidence.

Step 4: Collection

Acquire evidence.

Step 5: Examination

Extract information.

Step 6: Analysis

Determine events.

Step 7: Reporting

Prepare report.

Step 8: Presentation

Present evidence.

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10. Preparation

Definition

Preparation means planning and readiness before incidents occur.

Activities

- Train staff.
- Install forensic tools.
- Develop procedures.
- Create response teams.

Real-Time Example

A bank prepares for cyber attacks by:

- Training SOC analysts.
- Maintaining incident response plans.
- Deploying SIEM systems.

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11. What To Do Before The Incident

Organizations should prepare before cyber attacks happen.

Checklist

Create security policies

Regular backups

Enable logging

Install IDS/IPS

Maintain incident response plan

Train employees

Conduct security audits

Patch systems

Real-Time Example

A company regularly backs up data.

When ransomware attacks, they restore files from backups instead of paying ransom.

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12. Incident Response Plan (IRP)

Definition

A documented procedure for responding to cyber security incidents.

Phases

1. Preparation

Prepare resources.

2. Detection

Identify attack.

3. Containment

Stop spread.

4. Eradication

Remove threat.

5. Recovery

Restore services.

6. Lessons Learned

Improve security.

Real-Time Example

Ransomware Attack

```
Detection

Disconnect infected PC

Remove malware

Restore backups

Review incident
```

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13. Incident Response Team (IRT)

Definition

A group responsible for handling security incidents.

Team Members

Role	Responsibility
Incident Manager	Leads investigation
Security Analyst	Analyzes attack
Forensic Investigator	Collects evidence
System Administrator	Restores systems
Legal Advisor	Handles legal issues
HR Representative	Employee-related issues

Real-Time Example

A company experiences data theft.

IRT:

- Detects attack.
- Isolates affected systems.
- Investigates evidence.
- Reports to management.

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14. Detecting Incidents

Definition

The process of identifying abnormal or malicious activities.

Detection Methods

1. Log Analysis

Example:

Repeated failed SSH logins.

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2. IDS/IPS Alerts

Example:

Snort detects port scanning.

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3. SIEM Monitoring

Example:

ELK Stack detects suspicious login attempts.

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4. Antivirus Alerts

Example:

Windows Defender detects malware.

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5. User Reports

Example:

Employee reports unusual emails.

Real-Time Example

An attacker attempts SSH brute-force attacks.

Logs show:

```
Failed password for root
Failed password for root
Failed password for root
```

Incident detected and investigated.

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15. Chain of Custody

Definition

Chain of Custody is the chronological documentation showing the collection, handling, transfer, and storage of evidence.

It proves that evidence has not been altered.

Why Important?

Without Chain of Custody:

Evidence may be rejected in court.

With Chain of Custody:

Evidence is legally admissible.

Information Recorded

- Evidence ID
- Description
- Date & Time
- Collected By
- Location
- Transfer Details
- Hash Values

Example Chain of Custody Form

Evidence ID	Description	Collected By	Date
HDD-001	Dell Laptop HDD	John Smith	10-Jan-2026
USB-002	Kingston USB	John Smith	10-Jan-2026

Real-Time Example

Police seize a suspect's laptop.

```
Crime Scene

Investigator

Forensic Lab

Court
```

Every transfer is documented and signed.
